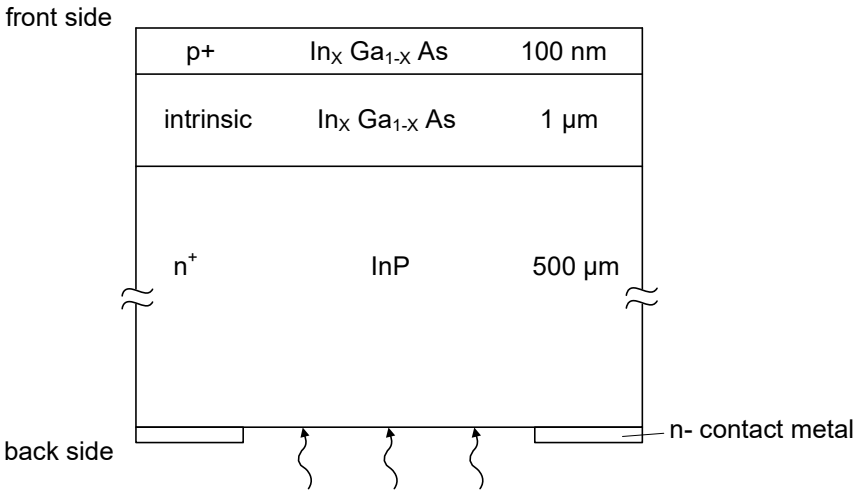
Advanced Optical Fiber Communications

Problem 15: Back illuminated InGaAs photodiode

The vertical structure of an InGaAs PIN photodiode on an InP substrate is shown in the figure below. The device is designed for illumination from the backside.



a) For lattice matched growth, the In content is fixed to $x = 0.53$. The bandgap energy is then $E_{g,InGaAs} = 0.75$ eV and the refractive index is $n_{InGaAs} = 3.56$. The InP substrate has a bandgap energy of $E_{g,InP} = 1.35$ eV and a refractive index of $n_{InP} = 3.4$. For which wavelength range can the back illuminated photodetector be used?

b) Determine the optical power entering the intrinsic absorption region from the backside for a wavelength of 1300 nm for vertical incident light from free space with an optical power of $P_0 = 10$ µW. Assume $2\alpha_{InP} = 0.25$ cm^{-1}.

c) Determine the absorbed optical power in the intrinsic absorption region for a single pass ($2\alpha_{InGaAs} = 5 \cdot 10^4$ cm^{-1}).

d) What is the transmitted optical power at the top interface, if the p contact is replaced by a thick metal contact with $R_{Metal} = 0.98$. Determine the reflected optical power at the intrinsic layer/metal interface. Calculate the quantum efficiency η and the total photo current generated in the intrinsic absorption region considering first and second pass only.

e) Determine the RC time constant for a circular diode with a diameter of 20 µm. Assume $R_D + R = 50\ \Omega$.

f) Calculate the drift time for $v_n = 1 \cdot 10^5$ m/s and estimate the total small signal 3 dB-bandwidth of the detector with the top metal contact.

Problem 16: coherent detector

Derive the expression for the intensity of the combined signal and local oscillator optical fields on the surface of the photodetector that produces a photocurrent in a coherent optical receiver. The signal field is given by:

$E_s = A_s \cos[\omega_s t + \varphi_s(t)]$

and the local oscillator field is given by

$E_{LO} = A_{LO} \cos[\omega_{LO} t + \varphi_{LO}(t)]$